

A Power Series Solution of the 't Hooft-Polyakov Monopole.

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The study of extended solutions of classical nonlinear field equations recently aroused a wide interest because of its relevance in problems of hadron physics.

Of particular interest is the magnetic-monopole solution obtained by 't Hooft ⁽¹⁾ and POLYAKOV ⁽²⁾ from an O_3 gauge theory with a triplet of scalars. Their result was then extended to include electrically as well as magnetically charged « dyons » ⁽³⁾ and also SU_3 and SU_4 solutions were studied ⁽⁴⁾.

However it has not been possible to find general analytical solutions to this problem, except for some artificial limits ⁽⁵⁾.

It is the purpose of this note to show that, with the aid of conservation laws one can easily decouple the equations of motion ⁽⁶⁾ corresponding to the 't Hooft-Polyakov model, thus obtaining analytical solutions in the form of series expansions and asymptotic behaviours of vector and scalar fields.

We start from the Lagrangian density

$$(1) \quad \mathcal{L} = -\frac{1}{4} F_{\mu\nu}^a F^{\mu\nu a} + \frac{1}{2} \pi_\mu^a \pi^{\mu a} + \frac{1}{2} \mu^2 \phi^a \phi^a - \frac{1}{4} \lambda (\phi^a \phi^a)^2,$$

where

$$F_{\mu\nu}^a = \partial_\mu A_\nu^a - \partial_\nu A_\mu^a + e\epsilon^{abc} A_\mu^b A_\nu^c, \quad \pi_\mu^a = \partial_\mu \phi^a + e\epsilon^{abc} A_\mu^b \phi^c.$$

We shall look for solutions satisfying: a) time independence, b) separability into radial (r) and angular (θ, φ) co-ordinates and c) finite energy.

Under these assumptions it has been shown ⁽⁷⁾ that the Wu-Yang-'t Hooft-Polyakov ansatz is unique and corresponds to a solution having the properties of a magnetic

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monopole of unit strength. We thus seek for a solution of this form, that is

$$(2) \quad \begin{cases} A_i^a = \varepsilon_{a i j} \check{r}_j (1 - K(r)) / e r, \\ A_0^a = 0, \\ \phi^a = \check{r}_a H(r) / e r. \end{cases}$$

Then the equations of motion reduce to

$$(3) \quad \begin{cases} r^2 K'' = K(K^2 - 1) + K H^2, \\ r^2 H'' = 2 H K^2 - \mu^2 r^2 H + (\lambda/e^2) H^3. \end{cases}$$

The appropriate boundary conditions for finite-energy solutions are

$$(4) \quad \begin{cases} K(0) = 1, & K(r) \xrightarrow{r \rightarrow \infty} 0, \\ H(0) = 0, & H(r)/r \xrightarrow{r \rightarrow \infty} \mu e / \lambda^{1/2}. \end{cases}$$

We now calculate the energy momentum and angular-momentum tensors, $T_{\mu\nu}$ and $M_{\lambda,\mu\nu}$, from the expressions

$$(5) \quad \begin{cases} T_{\mu\nu} = 2 \frac{\delta \mathcal{L}}{\delta g^{\mu\nu}} - \mathcal{L} g^{\mu\nu}, \\ M_{\lambda,\mu\nu} = x_\mu T_{\lambda\nu} - x_\nu T_{\lambda\mu} + F_{\lambda\mu}^a A_\nu^a - F_{\lambda\nu}^a A_\mu^a. \end{cases}$$

The only nonvanishing components of $T_{\mu\nu}$ are

$$(6) \quad \begin{cases} e^2 r^2 T_{rr} = [K'^2 - (1 - K^2)^2 / 2r^2 + (rH' - H)^2 / 2r^2 + \mu^2 H^2 / 2 - \lambda H^4 / 4e^2 r^2 - \alpha r^2], \\ 2e^2 r^2 T_{\varphi\varphi} = [(1 - K^2)^2 - (rH' - H)^2 + \mu^2 H^2 r^2 - \lambda H^4 / 2e^2 - 2\alpha r^4] \sin^2 \theta, \\ T_{\theta\theta} = T_{\varphi\varphi} / \sin^2 \theta, \\ T_{00} = -\mathcal{L} = \\ = [(K^2 - 1)^2 / 2 + H^2 K^2 + (rH' - H)^2 / 2 - \mu^2 H^2 r^2 / 2 + \lambda H^4 / 4e^2 + \alpha r^4] / e^2 r^2, \end{cases}$$

where $\alpha = \mu^4 r^3 / 4\lambda$. (We took advantage of the freedom in the definition of $T_{\mu\nu}$ in order to have $\lim_{r \rightarrow \infty} T_{\mu\nu} = 0$.)

For $M_{\lambda,\mu\nu}$, the nonvanishing components take the form

$$(7) \quad \begin{cases} M_{r,r\theta} = r^2 \sin^2 \theta \cos \theta T_{rr}, \\ M_{\theta,\theta r} = r T_{\theta\theta} - K'(1 - K)/e^2, \\ M_{\varphi,\varphi r} = r T_{\varphi\varphi} - K'(1 - K) \sin^2 \theta / e^2. \end{cases}$$

We now write the conservation law $\partial^\mu T_{\mu\nu} = 0$ in spherical co-ordinates.

$$(8a) \quad \frac{d}{dr} (r^2 T_{rr}) = (2/r) T_{\theta\theta},$$

$$(8b) \quad T_{\varphi\varphi} = \sin^2 \theta T_{\theta\theta}.$$

We see from eqs. (6) that eq. (8b) is automatically satisfied. We have another non-trivial equation arising from the conservation law $\partial^\mu M_{\mu,\nu\lambda} = 0$

$$(9) \quad \frac{d}{dr} (r^2 T_{rr}) = -r T_{rr} + (1/r) T_{\theta\theta}.$$

From eqs. (8a) and (9) we see that the relation $T_{\theta\theta} = -r^2 T_{rr}$ holds. This, together with the equations of motion allows us to decouple the equation satisfied by the vector field

$$(10) \quad x^2 f'' = (x^2 - 1)f + (1/\gamma - 1) \pm f[f^4 + 2x^2(f'^2 - f^2)]^{\frac{1}{2}}.$$

The scalar field is then given by the expression

$$(11) \quad H^2 = x^2 - f^2 \pm [f^4 + 2x^2(f'^2 - f^2)]^{\frac{1}{2}},$$

where $\gamma = e^2/\lambda$, $x = \mu r \gamma^{\frac{1}{2}}$ and $f = \gamma^{\frac{1}{2}} K$.

Equation (10) is rather complicated to look at. However, if one makes use of an auxiliary function u satisfying

$$(12) \quad (f^2 + u)^2 = f^4 + 2x^2(f'^2 - f^2),$$

one obtains

$$(13) \quad x^2 f'' = (x^2 - 1)f + (1/\gamma - 1 \pm 1)f^3 \pm fu,$$

$$(14) \quad H^2 = x^2 - (1 \mp 1)f^2 \pm u.$$

If we now choose $\gamma = \frac{1}{2}$ (which corresponds to $\lambda/e^2 = 2$) and the lower sign, we arrive to the simpler system

$$(15) \quad \begin{cases} x^2 f'' = (x^2 - 1)f - fu, \\ x^2(f'^2 - f^2) = f^2 u + u^2/2, \\ H^2 = x^2 - 2f^2 - u. \end{cases}$$

We then concentrate on the above mentioned particular relation between coupling constants and try to solve eqs. (15). We seek for an even expansion of the form

$$(16) \quad f = \sum_{n=1}^{\infty} c_n x^{2n-2}, \quad u = \sum_{n=1}^{\infty} d_n x^{2n-2}.$$

The boundary conditions for functions f and u are

$$(17) \quad f(0) = 1/\sqrt{2}, \quad f(x) \xrightarrow{x \rightarrow \infty} 0,$$

$$(18) \quad u(0) = -1, \quad u(x) \xrightarrow{x \rightarrow \infty} 0.$$

Condition (17) allows us to determine $c_1 = 1/\sqrt{2}$ and $d_1 = -1$. If we are able to fix c_2 and d_2 from the conditions at infinity, then all coefficients in expansion (16) will be

determined from the recurrence relations

$$c_{n+1} = \left(c_n - c_1 d_{n+1} - \sum_{m=2}^n c_m d_{n+2-m} \right) / [2n(n-1)],$$

$$d_{n+1} = 2 \left(c_1(c_n - c_{n+1}) + \sum_{m=2}^n \left\{ c_1 c_m + \frac{1}{2} d_m \right\} d_{n+2-m} + c_m c_{n+1-m} - \right. \\ \left. - 4(m-1)(n+1-m) c_m c_{n+2-m} + \sum_{k=1}^m c_{n+2-m} c_k d_{n+1-k} \right).$$

To fix c_2 and d_2 we use series (16) together with its analytic prolongations in order to be able to impose conditions at infinity. We have done this numerically, obtaining for c_2 and d_2 :

$$(19) \quad c_2 = -0.423 \dots, \quad d_2 = 2.20 \dots$$

Using these values and the recurrence relation one can obtain all the coefficients in expansions (16). We give in fig. 1 a schematic representation of the behaviour of the radial dependence on the fields.

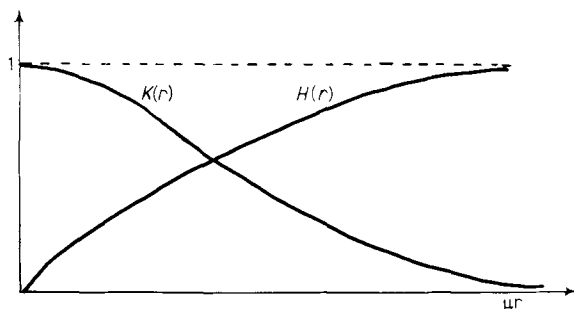


Fig. 1. - Schematic representation of the behaviour of $H(r)$ and $K(r)$. The functions $H(r)$ and $K(r)$ correspond, respectively, to the radial dependence on the Higgs field and on the vector field.

We have also studied the asymptotic behaviour of the solution with the aid of the Green's function method. We established the formula

$$(20) \quad f = Z \exp[-x] - \frac{1}{2} \exp[-x] \int_x^\infty \exp[y] p(y) dy + \frac{1}{2} \exp[x] \int_x^\infty \exp[-y] p(y) dy,$$

where

$$(21) \quad Z = \frac{1}{2} \sqrt{2} + \frac{1}{2} \int_0^\infty \exp[y] p(y) dy, \quad p(y) = (1+u) f/y^2.$$

From the preceding equations one can easily obtain the asymptotic behaviour of the vector potential

$$(22) \quad K \simeq \sqrt{2} Z \exp[-x] \left(1 - \frac{1}{2x} + \frac{1}{4x^2} - \frac{1}{4x^3} + O(1/x^5) \right) + O(\exp[-3x/x^2]).$$

The asymptotic form of eqs. (3) gives then the general solution $H = x + h$, where h vanishes as $\exp[-2x]$ at infinity.

By integrating T_{00} we are able to determine the value of the energy density. We have done this numerically, obtaining

$$(23) \quad E = \frac{\mu}{\alpha \sqrt{2}} f(\gamma), \quad f(\gamma) = 1.43 \dots,$$

where $\alpha = e^2/4\pi$. This result corresponds, as it was pointed out, to the particular choice $\lambda/e^2 = 2$, and is in good accord with the value obtained for f by 'T HOOFT. In his calculations, f varies from $f = 1.1$ for $\lambda/e^2 = 0.05$ to $f = 1.42$ for $\lambda/e^2 = 5$.

As a first application of the solution we have calculated the scalar- and vector-field form factors in the classical limit ⁽⁸⁾

$$(24) \quad \left\{ \begin{aligned} \langle \mathbf{p} | \hat{\phi}^a(0) | \mathbf{p}' \rangle &= \int \frac{d^3y}{4\pi} \exp[-i\mathbf{y} \cdot (\mathbf{p} - \mathbf{p}')] \phi^a(y) = \varepsilon_{12a} F(|\mathbf{q}|), \\ \langle \mathbf{p} | A_i^a(0) | \mathbf{p}' \rangle &= \int \frac{d^3y}{4\pi} \exp[-i\mathbf{y} \cdot (\mathbf{p} - \mathbf{p}')] A_i^a(y) = \varepsilon_{a13} G(|\mathbf{q}|), \end{aligned} \right.$$

where

$$(25) \quad \left\{ \begin{aligned} F(|\mathbf{q}|) &= (1/e) \int_0^\infty H(r) [\cos |\mathbf{q}|r/|\mathbf{q}| - \sin |q|r/|q|^2] dr, \\ G(|\mathbf{q}|) &= (1/e) \int_0^\infty (1 - K(r)) [\cos |\mathbf{q}|r/|\mathbf{q}| - \sin |q|r/|q|^2] dr. \end{aligned} \right.$$

Here $\mathbf{q} = \mathbf{p} - \mathbf{p}'$ has been chosen along the z -direction and the ket $|\mathbf{p}\rangle$ represents a monopole state with momentum $\mathbf{p}$.

By Fourier transforming the asymptotic expressions for the fields one can study the form factor singularities. For the vector field one can easily see that the form factor has a pole at $t = -\mathbf{q}^2 = \mu^2/2$. The scalar field form factor has a pole at $t = 0$ (which corresponds to the Goldstone boson) and another at $t = -2\mu^2$ ^(*).

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^(*) The field masses for the case $\lambda/e^2 = 2$ are: $M_\omega^2 = \mu^2/2$ and $M_\phi^2 = 2\mu^2$.